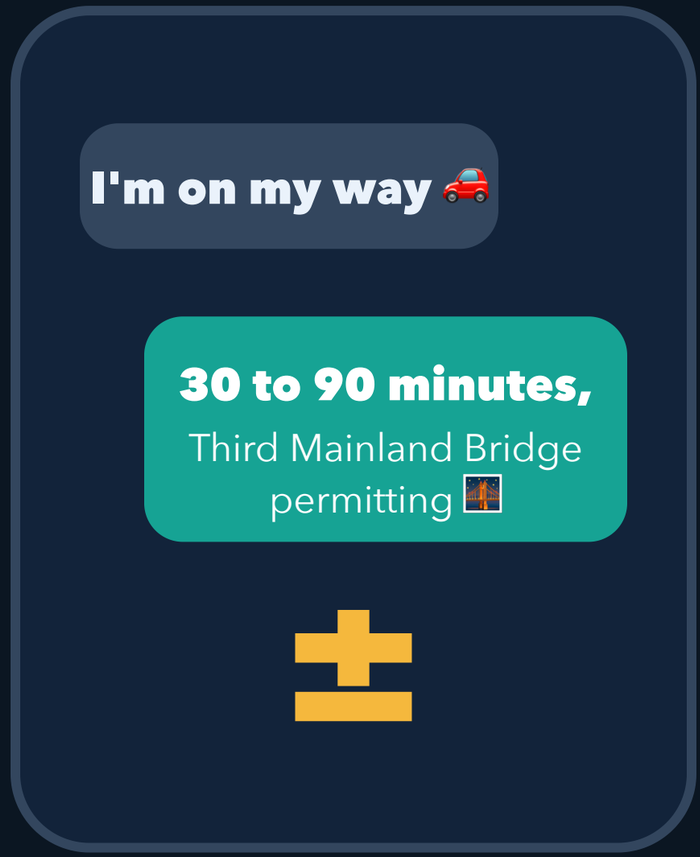
WEEK 5 · DAY 3

# Regression Forecasts + Honest Ranges

A forecast is a guess. An honest forecast says how wrong it might be, and checks that it was right. ⚖️




NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

# One rule, one fair fight, one honest range




⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 50 min hands-on · 🏁 20 min Checkpoint 4 · 🎬 10 min wrap-up

QUICK QUESTION

# Your friend is coming from the Island. 📱

🚗

**Tobi: “I'm on my way”**

One number, no warning. Could be 10 minutes. Could be 2 hours. You find out when the food is cold.

**point forecast**



**Amaka: “30 to 90 minutes, Third Mainland permitting”**

A most-likely time, a range, and the thing that could blow it up.

**interval forecast**

You can **plan** around Amaka. And if she says “90% of the time” she'd better be right 9 times in 10, not 5. 😏

LAST TIME · THE 8 HELD-OUT WEEKS (3 NOV – 22 DEC 2025)

# The bar any model must clear




MAE, tickets per week. Grandma + the bigger network: **2.24**, all 8 weeks forecast blind.

PART 1

# A forecast is a regression with one rule 🔒

THE LEAK RULE

# Every feature must be known before the week starts




`shift(1)`: on 3 Nov's row you see **27 Oct's 74**. You knew that on Monday morning. ✅

`shift(−1)`: you see **10 Nov's 54**. That's reading **tomorrow's newspaper**. 📰❌

THE CODE BOOK'S FEATURE TABLE · MAKE_FEATURES(TICKETS)

# Five columns the model reads

⏮ **tickets_lag1**
last week's count: `shift(1)`




**tickets_lag52**
the same week last year: `shift(52)`



**terminals**
how big the network is this week



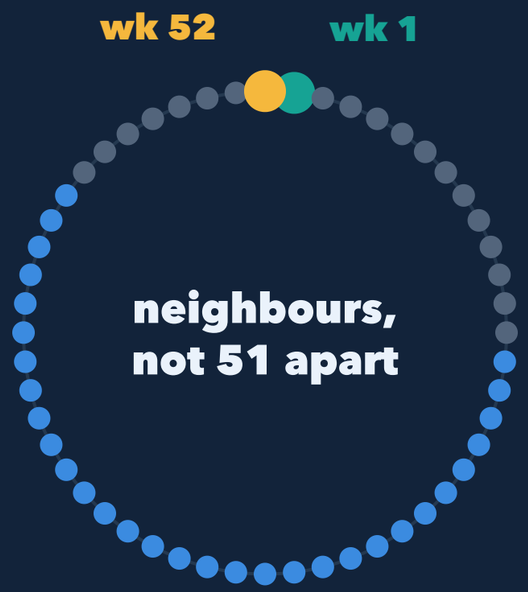
**sin52 + cos52**
week of the year on a clock (next slide)



**target**
this week's tickets: the answer the regression learns to reproduce

WHY SIN AND COS?

# The year is a clock, not a ruler 🕰️



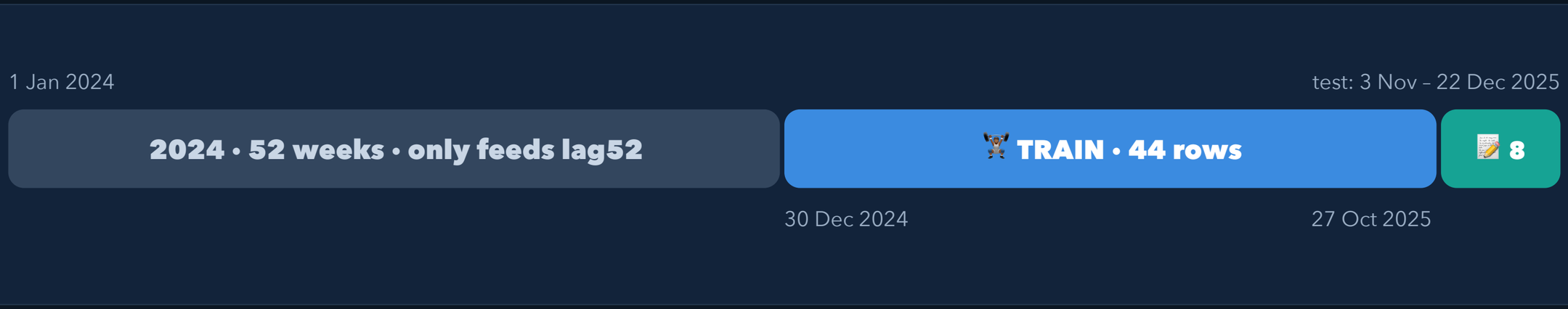

As a plain number, week 52 and week 1 look **51 apart**. On the calendar they are next-door neighbours.

`sin52` and `cos52` are where the week sits on the circle: **two columns instead of 52**, and the cycle joins up.

```
woy = df.index.isocalendar().week
f["sin52"] = np.sin(2 * np.pi * woy / 52)
f["cos52"] = np.cos(2 * np.pi * woy / 52)
```

WHERE THE ROWS GO

# 104 weeks → 52 usable rows




`shift(52)` has nothing to copy in 2024, so `dropna()` removes the first **52 rows**.

Train on **44 weeks**, test on the same **8 weeks** as last time. Five columns, 44 rows: a small exam.

FIT, THEN READ THE COEFFICIENTS LIKE AN ENGINEER

# Does every sign make sense? 🧐

```
reg = LinearRegression().fit(
    train_f[X_COLS], train_f["target"])
pred = reg.predict(test_f[X_COLS])
```

`terminals` **+0.0231** per terminal = about **+23 tickets a week per 1,000 terminals**. Bigger network, more tickets. ✅

A **negative** terminals sign would be a warning: investigate before you trust anything.

| COLUMN | COEFFICIENT |
|---|---|
| tickets_lag1 | +0.4974 |
| tickets_lag52 | +0.2259 |
| terminals | +0.0231 |
| sin52 | +0.8881 |
| cos52 | −2.9379 |

PART 2

# Like with like ⚖️

Same weeks, same horizon, same information.

THE REGRESSION'S SCORE ON THE 8 HELD-OUT WEEKS

# “Half of naive's error!” 🎉



**naive (W5D2)**

**11.25**

tickets/week off



**regression**

**5.85**

tickets/week off

Before you clap: **what did each one know** when it made each forecast? 🤔

THE HEAD START

# One sat the exam blind. One got hints. 🎁

🙈

**naive: 8 weeks, blind**

Stood on 27 Oct and said 74 for all 8 weeks. Never saw a single November count.

🎁

**regression: 1 week ahead**

`tickets_lag1` handed it **last week's REAL count** every single week. That's a one-week-ahead forecast.

Make naive the same way (last week's real count): **6.75** vs regression **5.85**. A win of about **0.9 ticket**, not half.

THE SAME 8 WEEKS · ONE SCORER (MAE) · SAY HOW FAR AHEAD

# Beat the BEST baseline, not the weakest

| METHOD | HOW FAR AHEAD | INFORMATION USED | MAE |
|---|---|---|---|
| **naive (W5D2)** | 8 weeks, blind | the last training week | **11.25** |
| **seasonal naive (W5D2)** | 8 weeks, blind | the same weeks last year | **15.25** |
| **one-step naive** | 1 week | last week's REAL count | **6.75** |
| **regression** | 1 week | last week's REAL count + 4 more columns | **5.85** |
| **rate seasonal naive (W5D2)** | 8 weeks, blind | last year's tickets per terminal × terminals | **2.24** |

Even with a hint every week, the regression (5.85) loses to a forecast made **8 weeks blind** (2.24). 🏆👵🏾

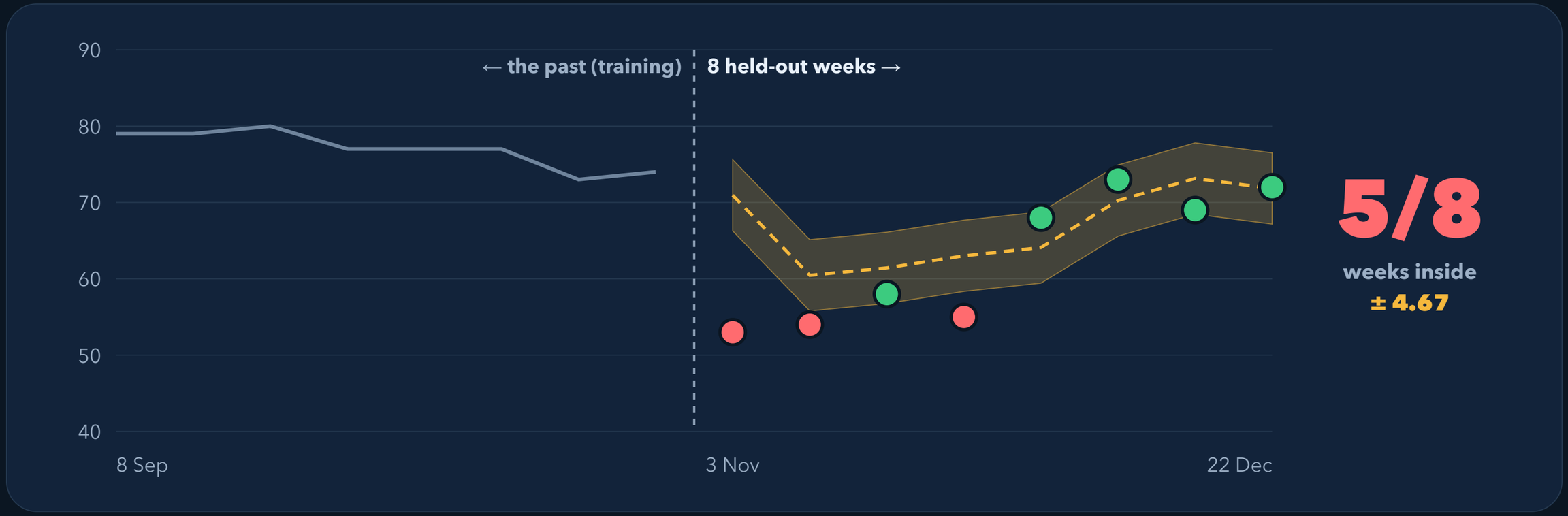
PART 3

# Honest ranges ↔️

Amaka's sentence, with a number on it, checked.

THE CODE BOOK'S 90% BAND · FORECAST ± 1.645 × RESIDUAL SD (2.84) = ± 4.67

# A “90%” band that's right 5 times in 8 😬




🟢 inside the band · 🔴 outside · gold dashes = the regression's forecast. **62.5%** is not 90%.

WHY SO NARROW?

# Two reasons, both fixable 🔍

🏠

### It marked its own homework

Training misses: sd **2.84**. Misses on weeks it never saw: sd **6.99**, and on average 4.1 tickets **too high**.

🌧️➡️☀️

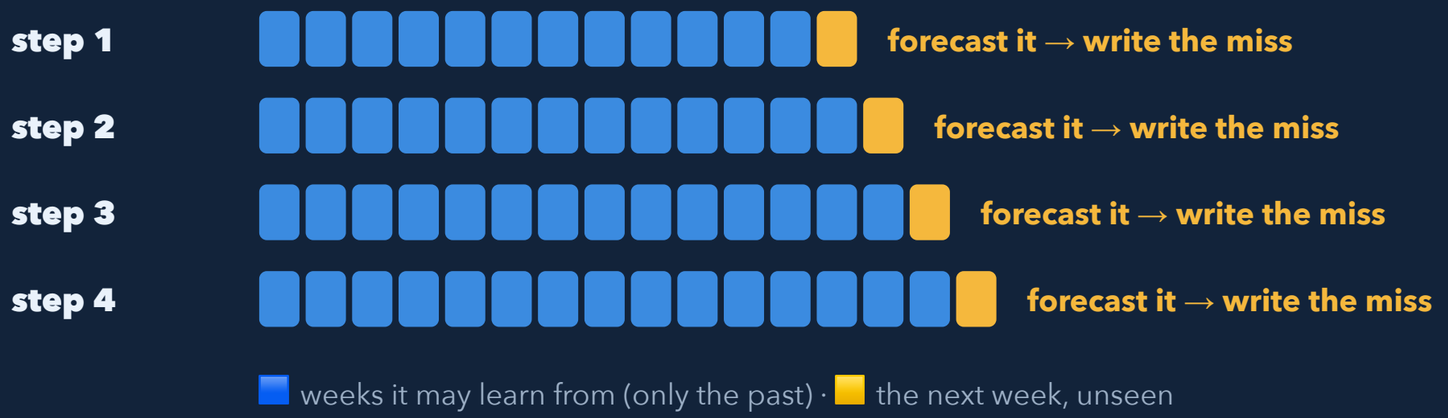
### The season switched off

3 Nov: the rains end. Last week was 74, it forecast **70.9**, reality was **53** (-17.9). A smooth sin/cos wave can't make a step.

A band measured on easy weeks is **confident about the wrong things**.

THE HONEST RECIPE

# Measure the misses it made before it knew the answer

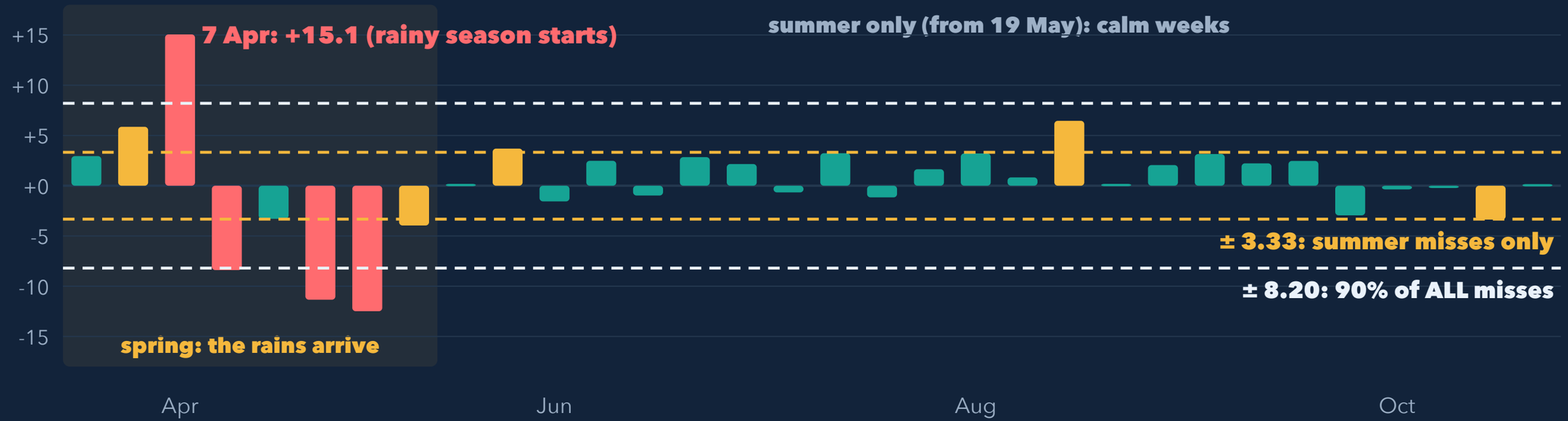



Walk forward through the training weeks from **24 Mar**: fit, forecast the next week, write down the miss. **32 honest misses.**

Band = forecast ± the **90th-percentile miss**. No bell-curve guess: just how wrong it actually was.

THE REGRESSION'S 32 HONEST PAST MISSES · ONE WEEK AHEAD

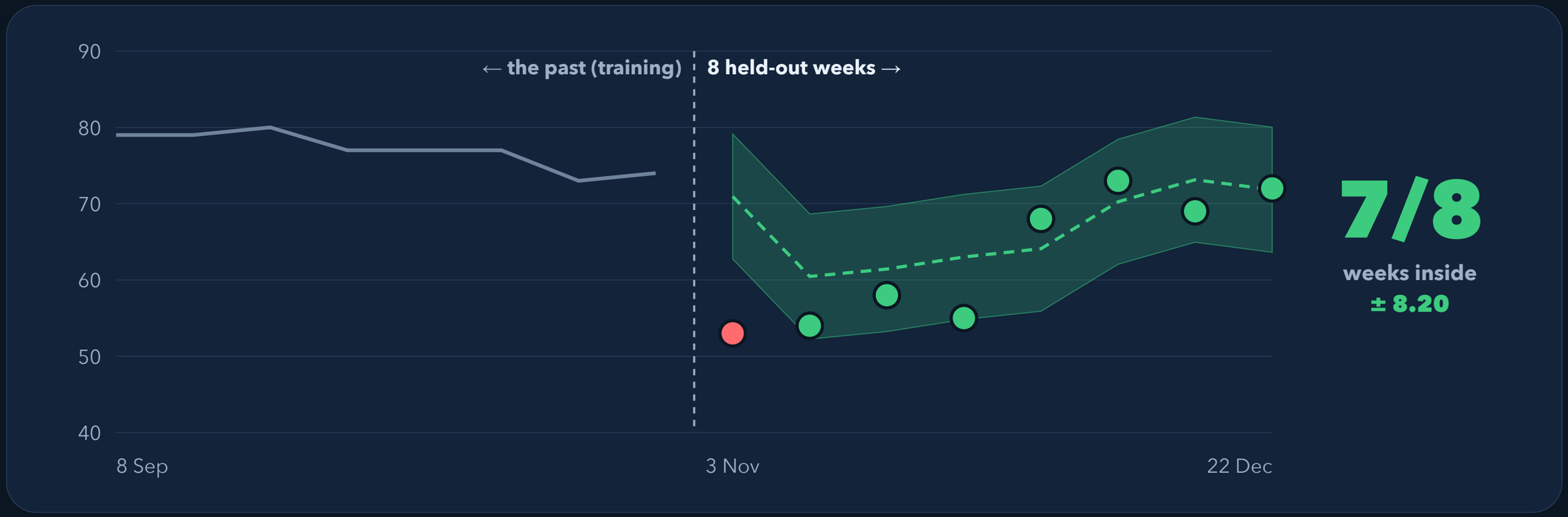
# The big misses live at the season change




Include the spring weeks: **± 8.20**. Use summer only: **± 3.33**.

SAME FORECAST · BAND FROM THE HONEST MISSES · ± 8.20

# 7 of 8. That's a band you can quote ✅




The one miss is the season switch, 3 Nov. Same forecast as before: **only the width changed**.

THE TRAP

# A band is only as honest as the weeks you measured it on



**Misses from 24 Mar**

includes the April switch

**7/8**

± 8.20



**Misses from 19 May**

summer only: calm weeks

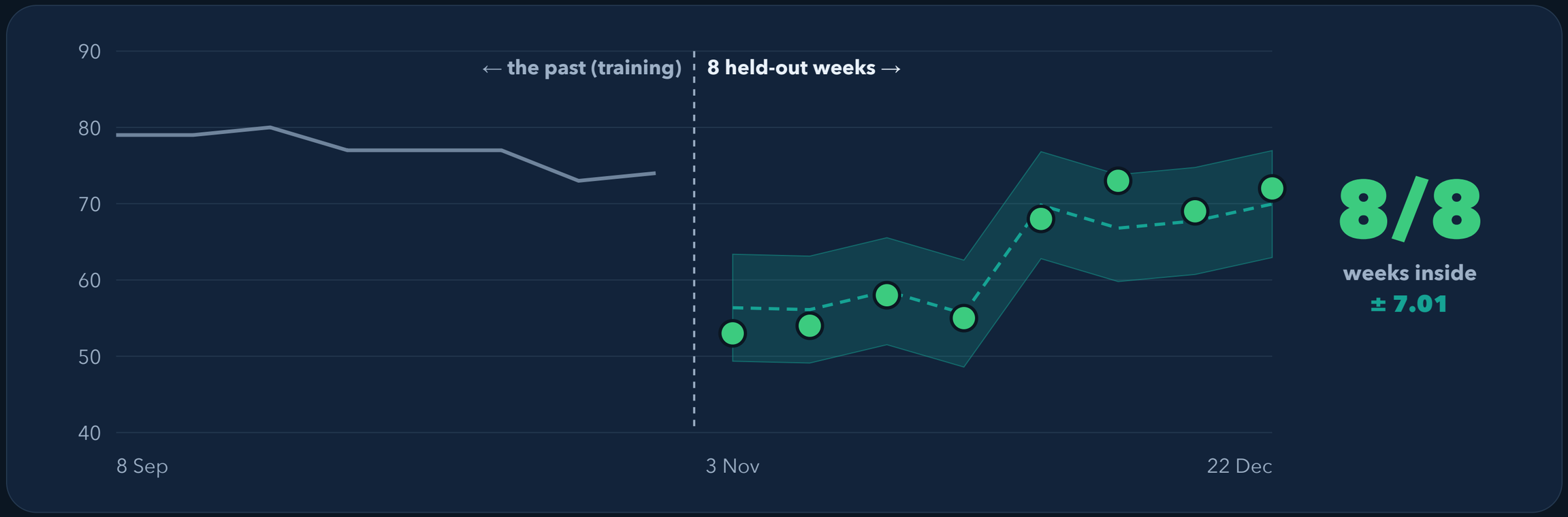
**2/8**

± 3.33

You're forecasting a season change? **Measure on season changes.**

GRANDMA + THE BIGGER NETWORK, WITH AN HONEST BAND · ± 7.01

# The best forecast gets a band too 🏆




Misses of every 8-week rate forecast made from **37** earlier starting weeks (296 misses) → **8 of 8** inside.

TWO HABITS

# Don't turn the knob. Say it like Amaka.



### Raising z until the test looks right

That's tuning on the answers: **expo again**. The width must come from misses **before** the test.



### The honest sentence

“Week of 3 Nov: most likely around **56**, plausibly **49 to 63**; expect bigger misses at season changes.”

Most likely · a range · the thing that could blow it up. Three parts, every time.

🏁 FINAL PROJECT · CHECKPOINT 4

# ⚖️ Honest test: bring a table like this

| METHOD | HOW FAR AHEAD | MAE | 90% BAND | COVERAGE |
|---|---|---|---|---|
| **naive** | 8 weeks, blind | 11.25 | — | — |
| **seasonal naive** | 8 weeks, blind | 15.25 | — | — |
| **rate seasonal naive** | 8 weeks, blind | 2.24 | ± 7.01 | 8 of 8 |
| **one-step naive** | 1 week | 6.75 | — | — |
| **regression • code-book band** | 1 week | 5.85 | ± 4.67 | 5 of 8 |
| **regression • honest band** | 1 week | 5.85 | ± 8.20 | 7 of 8 |

✅ split by time + `assert` + embargo
✅ your baseline on the **same period, same scorer**

⛔ real data stays on NIGCOMSAT systems, not Colab
Next: **Week 6 • Day 3: 🎤 Demo day**

YOUR TURN · 50 MINUTES

# Forecast + fence ↔️

1. Open **W5D3-exercise**, save a copy, run STEP 0
- **A** Features without a leak + the split
- **B** Fit, read the signs, compare like with like
- **C** Three bands: which ones are honest?
- **D** Your Checkpoint 4 table



**14 checks to pass**

🌶️ Pepper task: write Amaka's sentence with real numbers

Keep the **rule book** open next to you.

TODAY IN 3 LINES

# What you now know

🔒 **Leak rule:** every feature known before the week starts. `shift(1)`, never `shift(−1)`.

⚖️ **Like with like:** same weeks, same horizon, same information. Beat the **best** baseline (2.24).

↔️ **Honest band:** width from past misses that include the hard weeks, then check the coverage.

Next 👉 **Backtesting**: one split of 8 weeks… is that enough? 🎰